

The 50+ Experiments: Computational Validation of the Cathedral via Certified Rust/MPFR/DD Pipelines

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Abstract

We document the 50+ companion experiments that validate the Cathedral’s formal proof architecture. Each experiment is implemented in Rust, using f64 through 512-bit MPFR and DD (31-digit Dekker–Knuth) arithmetic, and produces either numerical validation data or certified JSON output consumed by Lean oracle axioms. The experiments span six categories: Gram matrix computation, Mellin/Parseval validation, Perron contour integration, spectral analysis, character-spectral probes, and DD-precision certification. We describe the certification pipeline that bridges computational results to formal proofs, achieving Sherman–Morrison match to 10^{-17} , eigenvalue certification through $N = 1000$, and DD-precision Gram matrix certification at $N = 55,440$.

Contents

1	Introduction	1
2	Experiment Catalog	4
3	The Certification Pipeline	5
3.1	Architecture	5
3.2	Three Certificate Types	5
3.3	The Precision Wall and the Dekker–Knuth Solution	5
4	Key Experimental Results	5
4.1	Rayleigh Quotient Convergence	5
4.2	Báez-Duarte Distance Convergence	5
4.3	Parseval Bridge Validation	6
4.4	Universal Equation of State	6
4.5	The $\alpha \approx 0.47$ Constant	6
4.6	Torus Projection (June 2026)	6
4.7	Spectral Mirror (May 2026)	6
5	Conclusion	6

1 Introduction

The Cathedral’s formal proofs establish logical equivalences; the experiments validate that the mathematical objects behave as the theory predicts. This two-channel architecture—formal proof for logical structure, computation for numerical reality—provides defense in depth.

All experiments live in `experiments/` and are standalone Rust crates with `Cargo.toml` build files. Most use the `rug` crate (bindings to GMP/MPFR) for arbitrary-precision arithmetic.

2 Experiment Catalog

#	Name	Category	Precision
1	gram-matrix	Gram	512-bit MPFR
2	gram-oracle	Gram	Exact rational
3	gram-form-identity	Gram	256-bit MPFR
4	gram-bilinear-abel	Gram	256-bit MPFR
5	gram-pointwise	Gram	256-bit MPFR
6	gram-quadform	Gram	256-bit MPFR
7	mellin-certificate	Mellin	256-bit MPFR
8	contour-oracle	Mellin	f64
9	perron-contour	Perron	256-bit MPFR
10	mvt-decomposition	Perron	256-bit MPFR
11	crown-cancellation	Perron	256-bit MPFR
12	vasyunin	Vasyunin	Exact rational
13	vasyunin-convergence	Vasyunin	256-bit MPFR
14	vasyunin-integral	Vasyunin	256-bit MPFR
15	spectral	Spectral	f64
16	spectral-analyzer	Spectral	512-bit MPFR
17	hilbert-spectral	Spectral	256-bit MPFR
18	rotor-spectroscopy	Spectral	f64
19	two-tile-analyzer	Spectral	f64
20	van-hove-probe	Spectral	f64
21	character-spectral	Character	f64 + MPFR
22	baez-duarte	BD	512-bit MPFR
23	numerical	Numerical	f64
24	l2-decay-certificate	L^2	256-bit MPFR
25	norm-bound-validator	Bounds	256-bit MPFR
26	pnt-mobius-sums	PNT	f64
27	mobius-basis	PNT	f64
28	siegel-walfisz	PNT	f64
29	covariance-probe	Covariance	256-bit MPFR
30	fejer-kernel	Analysis	f64
31	abel-bridge	Abel	256-bit MPFR
32	abel-tail-validator	Abel	256-bit MPFR
33	algebraic	Algebraic	Exact
34	millennium-wall	Misc	f64
35	bc-exponent-frontier	Zeta	f64
36	bc-witness-analysis	Zeta	f64
37	bc-zeta-lower	Zeta	256-bit MPFR
38	cert-export	Certified	f64 + JSON
39	nb-witness-scan	BD	f64 (midpoint quad)
40	certified-distance	DD	DD + MPFR
41	cathedral-utils	Shared	f64 + MPFR
42	gram-scaling-oracle	Gram	256-bit MPFR
43	two-tile-decomposition	Analysis	f64
44	nb-witness-scan-gpu	BD (GPU)	f32/f64
45	renormalization-bridge	Renorm	f64
46	moebius-microscope	Gram (GPU)	MPFR-256 + CUDA
47	hc-gram-oracle	DD cert	DD + HPDF
48	spectral-observatory	Spectral	DD MPFR-256
49	prime-harmonics	Mirror	f64 (10K zeros)
50	overcancellation-scan	Analysis	f64 + MPFR
51	torus-projection	Torus	f64
52	selberg-ecot-exponent	Exponent	f64
53–55+	(additional experiments)	Various	Mixed

3 The Certification Pipeline

3.1 Architecture

The `cert-export` experiment is the bridge between computation and formal proof. It:

1. Computes eigenvalues, eigenvectors, and spectral statistics.
2. Exports results as machine-readable JSON certificates.
3. Generates Lean 4 oracle axiom files consumed by `Compute/OracleCertificates.lean`.

This architecture implements *artifact-backed formal verification*: the heavy, heuristic computation (Rust/MPFR, floating-point eigensolvers) is cleanly separated from the formal logic (Lean 4), with only machine-readable certificates crossing the boundary. The Lean side consumes these as isolated oracle axioms—explicitly marked, auditable, and never contaminating the pure proof chain. This ensures that floating-point roundoff cannot propagate into the formal logic.

3.2 Three Certificate Types

1. **Eigenvalue certificates:** $\lambda_{\min}(G_N) > 0$ at $N \in \{50, 100, 200, 300, 500, 750, 1000\}$.
2. **Eigenvector localization:** participation ratios and prime-vs-composite weight distributions.
3. **Residue class decomposition:** block-diagonal eigenvalues for $m \in \{3, 4, 5, 6, 7, 8, 10, 12\}$.

3.3 The Precision Wall and the Dekker–Knuth Solution

At $N \geq 750$, the Gram matrix condition number exceeds 10^6 , causing standard f64 eigensolvers to hallucinate negative eigenvalues. At $N > 40,000$, the condition number exceeds 10^{15} and *every* f64 eigensolver produces non-physical results: the matrix appears non-positive-definite despite being provably SPD.

The `certified-distance` experiment solves this with Dekker–Knuth Double-Double (DD) arithmetic, providing ~ 31 digits of precision at only $2\times$ the cost of f64. A DD-precision Conjugate Gradient solver computes d_N^2 at $N = 55,440$ ($\dim = 55,439$), the largest such computation in the literature. The DD matrix is built by `cathedral-utils` using MPFR at 256-bit precision, then splitting each entry into (hi, lo) f64 pairs.

4 Key Experimental Results

4.1 Rayleigh Quotient Convergence

The `gram-oracle` experiment confirms $Q_N / \ln N \rightarrow C$ where $C \approx 21.649$, validating the spectral bridge prediction.

4.2 Báez-Duarte Distance Convergence

The `nb-witness-scan` experiment ($N = 2$ to $20,000$, midpoint quadrature with $50,000$ points) confirms the scaling law $d_N^2 \approx C_{\text{holes}} / \ln N$:

N	d_N^2	$d_N^2 \cdot \ln N$	PNT sum S_1
100	1.331×10^{-1}	0.613	0.031
500	7.506×10^{-2}	0.467	-0.009
1000	5.959×10^{-2}	0.412	0.004
2000	5.012×10^{-2}	0.381	—
5000	4.003×10^{-2}	0.341	—
10000	3.439×10^{-2}	0.317	—
20000	3.073×10^{-2}	0.305	0.001

The product $d_N^2 \cdot \ln N$ is slowly decreasing, consistent with the Báez-Duarte prediction $d_N^2 \sim c_{\text{holes}} / \ln N$ where $c_{\text{holes}} = 2 + \gamma - \ln(4\pi) \approx 0.0462$. The PNT sum $S_1 = \sum_{k \leq N} \mu(k)/k \rightarrow 0$ is confirmed to $O(10^{-3})$ at $N = 20,000$.

Total computation time: 2,307 seconds (MacBook Pro, M-series, 10 cores).

4.3 Parseval Bridge Validation

The `mellin-certificate` experiment confirms the Parseval bridge identity to 6.6×10^{-6} relative error by computing the L^2 norm in both the spatial and frequency domains.

4.4 Universal Equation of State

The `character-spectral` suite (fast-probe, deep-probe, modulus-probe) establishes the universal thermalization threshold $N_c \approx 60 \cdot m / \varphi(m)$ across 8 moduli, proving that the mod-8 residue class structure is not privileged.

4.5 The $\alpha \approx 0.47$ Constant

Eigenvector analysis through $N = 1000$ reveals persistent partial localization at participation ratio $\alpha \approx 0.47$, identifying a new arithmetic-GOE constant characterizing the boundary between number-theoretic order and random matrix chaos.

4.6 Torus Projection (June 2026)

The `torus-projection` experiment decomposes the Gram energy $v^T G v$ by GCD strata onto the infinite-dimensional torus $T^\infty = \prod_p S_p^1$. Per-prime energy analysis shows that prime $p = 2$ carries 46–86% of stratum energy, with coherence decaying for larger primes. Zeta-zero phases equidistribute with Weyl discrepancy decreasing as N grows—consistent with GOE universality. The Lean 4 companion `TorusProjection.lean` proves the GCD partition theorem with zero sorry and zero axioms.

4.7 Spectral Mirror (May 2026)

The `prime-harmonics` experiment reconstructs $\pi(x)$ from zeta zeros via the explicit formula. With 10,000 zeros: exact at $\pi(10) = 4$, $\pi(50) = 15$, $\pi(200) = 46$. Eight modes (mirror, sweep, portrait, bench, hunt, fine_scan, full, democracy) explore different aspects of the prime-zero duality.

5 Conclusion

The 55+ experiments provide computational validation at every level of the Cathedral’s proof architecture, from the Gram matrix entries to the spectral gap to the Parseval bridge to the thermodynamic equation of state to the DD-precision Dekker–Knuth frontier. The certified

JSON pipeline ensures that computational discoveries are automatically available to the formal proof engine. The DD-precision pipeline extends certification to $N = 55,440$, beyond the reach of any standard eigensolver. The torus projection experiment (June 2026) extends the analysis to the per-prime structure of the Gram energy on T^∞ , confirming that primes act as independent circles in the infinite-dimensional torus.

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References

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